

Multiple Choice (10 marks)

1. What is the total binding energy and the binding energy per nucleon of the nucleus of ${}^{52}_{24}\text{Cr}$ which has a mass of 51.95699 amu?
 - (a) 4.0×10^8 eV and 7.7×10^6 eV
 - (b) 4.2×10^8 eV and 8.1×10^6 eV
 - (c) 4.6×10^8 eV and 8.9×10^7 eV
 - (d) 4.6×10^8 eV and 8.9×10^6 eV
 - (e) 5.0×10^8 eV and 9.6×10^6 eV
2. A car covers a distance of 30 miles in 1/2 hour. What is its speed in miles per hour and in feet per second?
 - (a) 30 mi per hr, 60 ft per sec
 - (b) 90 mi per hr, 132 ft per sec
 - (c) 75 mi per hr, 110 ft per sec
 - (d) 60 mi per hr, 55 ft per sec
 - (e) 45 mi per hr, 88 ft per sec
3. An object of mass 100 g is at rest. A net force of 2000 dynes is applied for 10 sec. What is the final velocity? How far will the object have moved in the 10-sec interval?
 - (a) 150 cm/sec and 7.5 m
 - (b) 350 cm/sec and 17.5 m
 - (c) 450 cm/sec and 22.5 m
 - (d) 250 cm/sec and 12.5 m
 - (e) 200 cm/sec and 10 m
4. What is the ratio of intensities of Rayleigh scattering for the two hydrogen lines $\lambda_1 = 410$ nm and $\lambda_2 = 656$ nm?
 - (a) 4.23:1
 - (b) 10.24:1
 - (c) 8.67:1
 - (d) 2.98:1
 - (e) 5.67:1
5. Why can't we see the Sun's corona except during total solar eclipses?

- (a) We can't see magnetic fields
- (b) The corona is blocked by the Earth's atmosphere
- (c) The corona is hidden by the Sun's photosphere
- (d) The corona is made up mostly of charged particles which aren't luminous.
- (e) The Sun's corona is actually invisible to the human eye

True / False (10 marks)

Answer True or False

- 6. True or False: When a branch is bent, the top side primarily experiences compressive stress as it is pushed downward.
- 7. True or False: A photon with a wavelength of 1600.0 nm can cause an electron in germanium to jump from the valence to the conduction band at 300 K.
- 8. True or False: The period of revolution for two double stars of solar mass separated by 4 light years is approximately 9×10^7 years.
- 9. True or False: An ocular focal length of 4 cm would produce an overall magnification of 100 x with a 10 x objective magnification.
- 10. True or False: In a longitudinal wave, the vibrations move along and parallel to the direction of wave propagation.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Calculate the specific gravity of a rectangular bar of iron with a mass of 320 grams and dimensions of $2 \times 2 \times 10$ centimeters. Discuss the implications of your result in the context of material properties.
- 12. Calculate the rotational kinetic energy of a flywheel with a mass of 30 kg and a radius of gyration of 2 m, rotating at 2.4 rev/sec. Discuss the implications of your findings.

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